Model A Fit to the Real Data

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1 Summary

Model A (one psychometric curve per group, with all sessions pooled) was fit to all 12,937 trials of the Sirius data (42 sessions, 4 groups). The main results:

- **The distractor lowers the PSE** by -4.19 μm [-4.96, -3.44] for the left hand and -6.07 μm [-6.85, -5.30] for the right hand.
- **The distractor raises** γ_h, the rate of "high" responses to the weakest stimuli, by about 0.04 for both hands.
- **The distractor raises the JND** for the left hand ($+0.83$ μm [0.27, 1.37]); for the right hand the HDI includes 0. It has no clear effect on γ_l.
- These effects agree closely with Model B. The PSE shift is the same in both models to within 0.1 μm.
- **Model B fits the data much better.** Leave-one-out cross-validation favors it by 164 (SE 19) in expected log predictive density. Within single sessions, near the threshold, the observed responses vary more than Model A predicts. Model A therefore describes these 42 sessions pooled together; to generalize to a new session, Model B's population-level estimates are the appropriate ones.

Every figure and number in this report is produced by `Main_Analysis_A.py`; the cell "Numbers for Fit_A_Report.tex" prints the numbers that the earlier cells do not. All intervals are 95% highest density intervals (HDIs). PSE and JND are in micrometres of stimulus amplitude: $\text{PSE}_{\mu m} = 28 + 14.63\,\text{PSE}$ and $\text{JND}_{\mu m} = 14.63\,\text{JND}$, where 28 and 14.63 are the mean and sd of the 8 amplitude levels.

2 Model and Fit

The model is the one described in the Model A convergence report. Each group g (left_uni, left_bi, right_uni, right_bi) has one psychometric curve,

$$\psi(x) = \gamma_{h,g} + (1 - \gamma_{h,g} - \gamma_{l,g})\ \text{logit}^{-1}(\beta_{0,g} + \beta_{1,g}\, x),$$

shared by all its trials regardless of session, with the priors $\beta_0 \sim \mathcal{N}(0, 6)$, $\beta_1 \sim \mathcal{N}(4, 2)$ and $\gamma_h, \gamma_l = 0.25\, w$ with $w \sim$ Beta(mean 0.2, sd 0.15). These priors predate the design-based priors of Model B. The lapse prior has mean 0.05, well above the lapse rates found here, but the data

dominate it: the posterior sd of each lapse rate is 5–27% of its prior sd, and the PSE and JND posteriors are far narrower than their priors (Figure 3, Appendix A).

The groups have 3,230–3,237 trials each. The fit used NUTS with 4 chains of 1000 tuning and 5000 kept draws (20,000 draws in total) and took 2.4 minutes.

2.1 Sampling diagnostics

There were no divergences, and the tree depth never exceeded 5. All 40 parameters have $\hat{R} = 1.000$, bulk ESS $\geq$ 15,213 and tail ESS $\geq$ 8,802. The chains overlap completely (Figure 1).

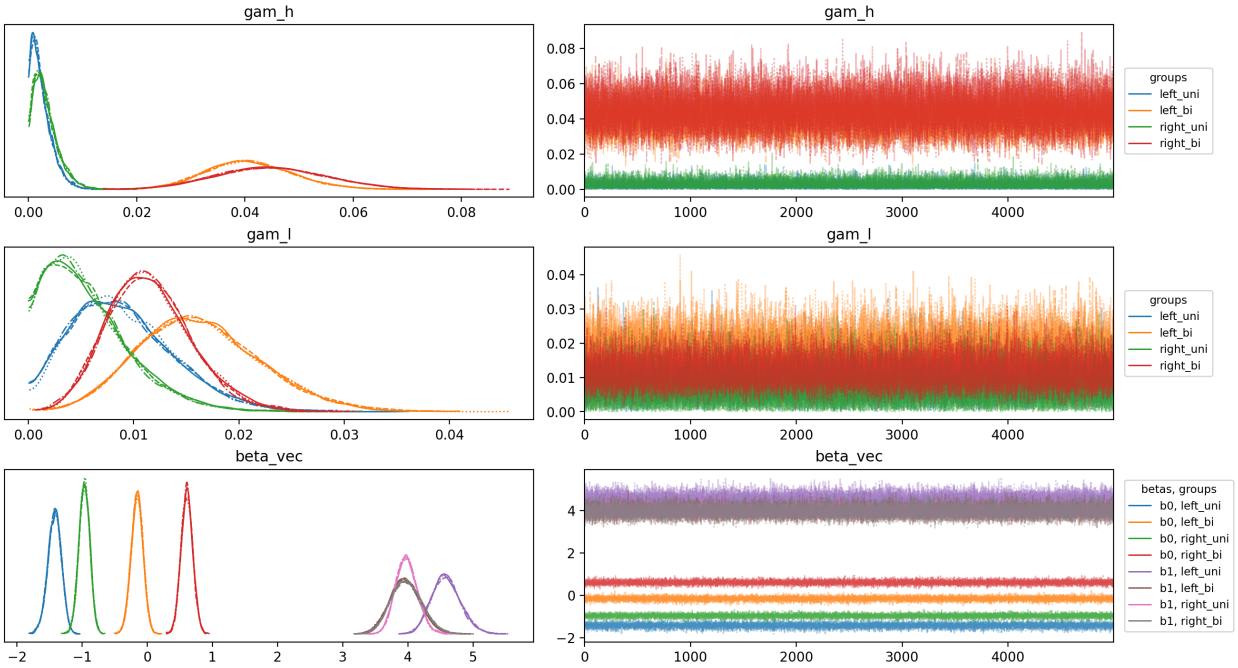


Figure 1: Trace plots of γ_h, γ_l, β_0 and β_1 for all groups (left: density per chain; right: draws).

3 Group Estimates

Table 1: Posterior mean [95% HDI] of the psychometric quantities of each group.

	PSE (μm)	JND (μm)	γ_h	γ_l
left_uni	32.58 [32.10, 33.04]	3.56 [3.25, 3.89]	0.002 [0.000, 0.006]	0.009 [0.000, 0.018]
left_bi	28.38 [27.79, 28.98]	4.39 [3.93, 4.83]	0.041 [0.025, 0.056]	0.016 [0.005, 0.027]
right_uni	31.57 [31.05, 32.06]	4.09 [3.77, 4.41]	0.003 [0.000, 0.008]	0.006 [0.000, 0.013]
right_bi	25.50 [24.89, 26.08]	4.36 [3.93, 4.81]	0.045 [0.025, 0.064]	0.012 [0.004, 0.020]

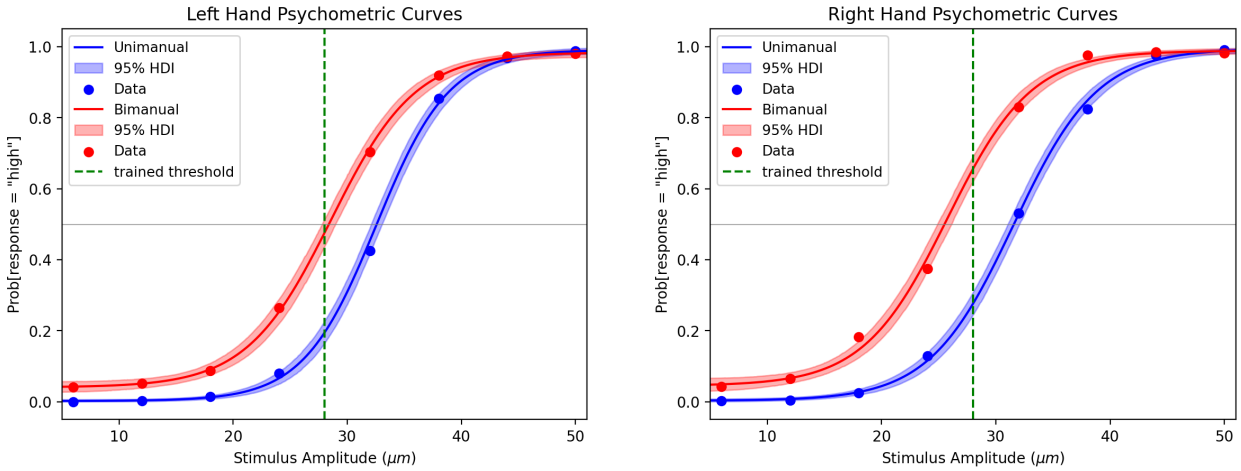


Figure 2: Fitted psychometric curves (at the posterior means of the parameters, with the pointwise 95% HDI of the curve) and the observed proportions of "high" responses, pooled over sessions. Unimanual in blue, bimanual in red; the dashed line is the trained threshold of 28 μm.

Without the distractor the PSE lies 4.6 (left) and 3.6 (right) μm above the trained threshold: the monkey needs a slightly stronger stimulus than the trained boundary before it reports "high". With the distractor the PSE moves to the threshold (left hand) or below it (right hand). Lapse rates are small; γ_h is close to 0 without the distractor and about 0.04 with it.

The Model A convergence study found that, at this data size and when the model is correct, the PSE, JND and γ_h are recovered without bias and with calibrated HDIs, while small values of γ_l are pulled upward by the prior. The γ_l estimates of 0.006–0.016 are therefore upper-leaning.

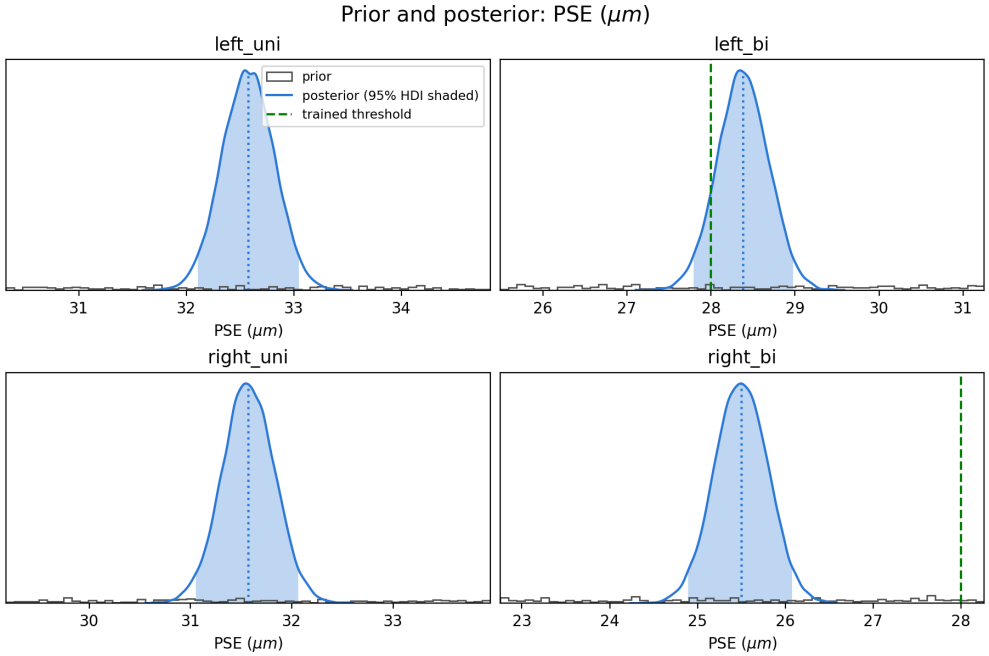


Figure 3: Prior (gray) and posterior (blue) of the PSE in each group.

4 Effect of the Distractor

The effect of the distractor is the bimanual − unimanual difference, computed draw by draw so that its uncertainty is exact.

Table 2: Distractor effect (bimanual − unimanual) under Model A: posterior mean [95% HDI] and the posterior probability that the effect is positive.

Hand	Quantity	effect	$P(\text{effect} > 0)$
Left	PSE (μm)	−4.19 [−4.96, −3.44]	0.000
	JND (μm)	0.83 [0.27, 1.37]	0.998
	γ_h	0.038 [0.023, 0.055]	1.000
	γ_l	0.007 [−0.008, 0.022]	0.832
Right	PSE (μm)	−6.07 [−6.85, −5.30]	0.000
	JND (μm)	0.27 [−0.29, 0.80]	0.837
	γ_h	0.042 [0.021, 0.061]	1.000
	γ_l	0.006 [−0.006, 0.017]	0.858

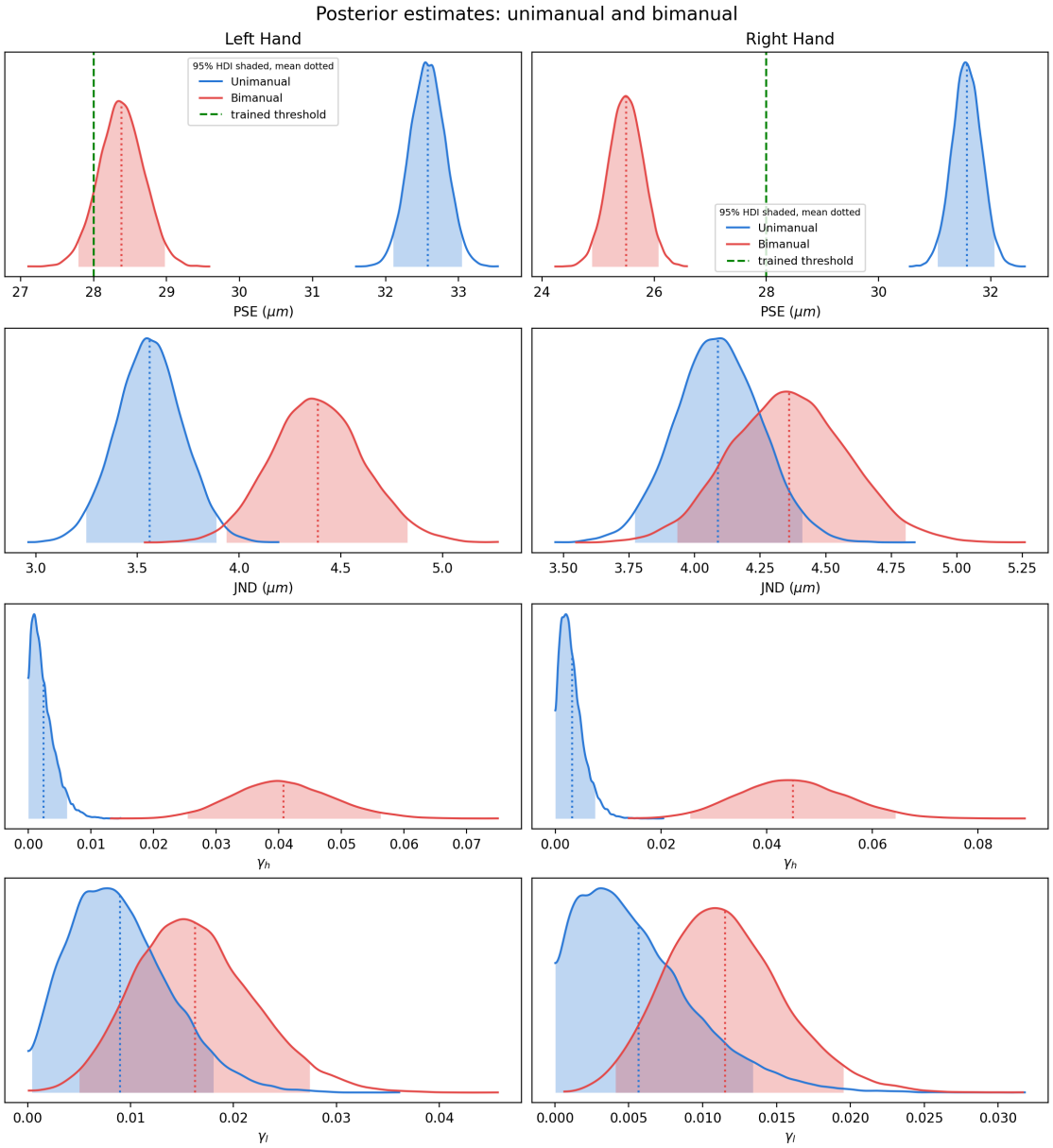


Figure 4: Posteriors of PSE, JND, γ_h and γ_l for the unimanual (blue) and bimanual (red) condition of each hand, with 95% HDIs shaded.

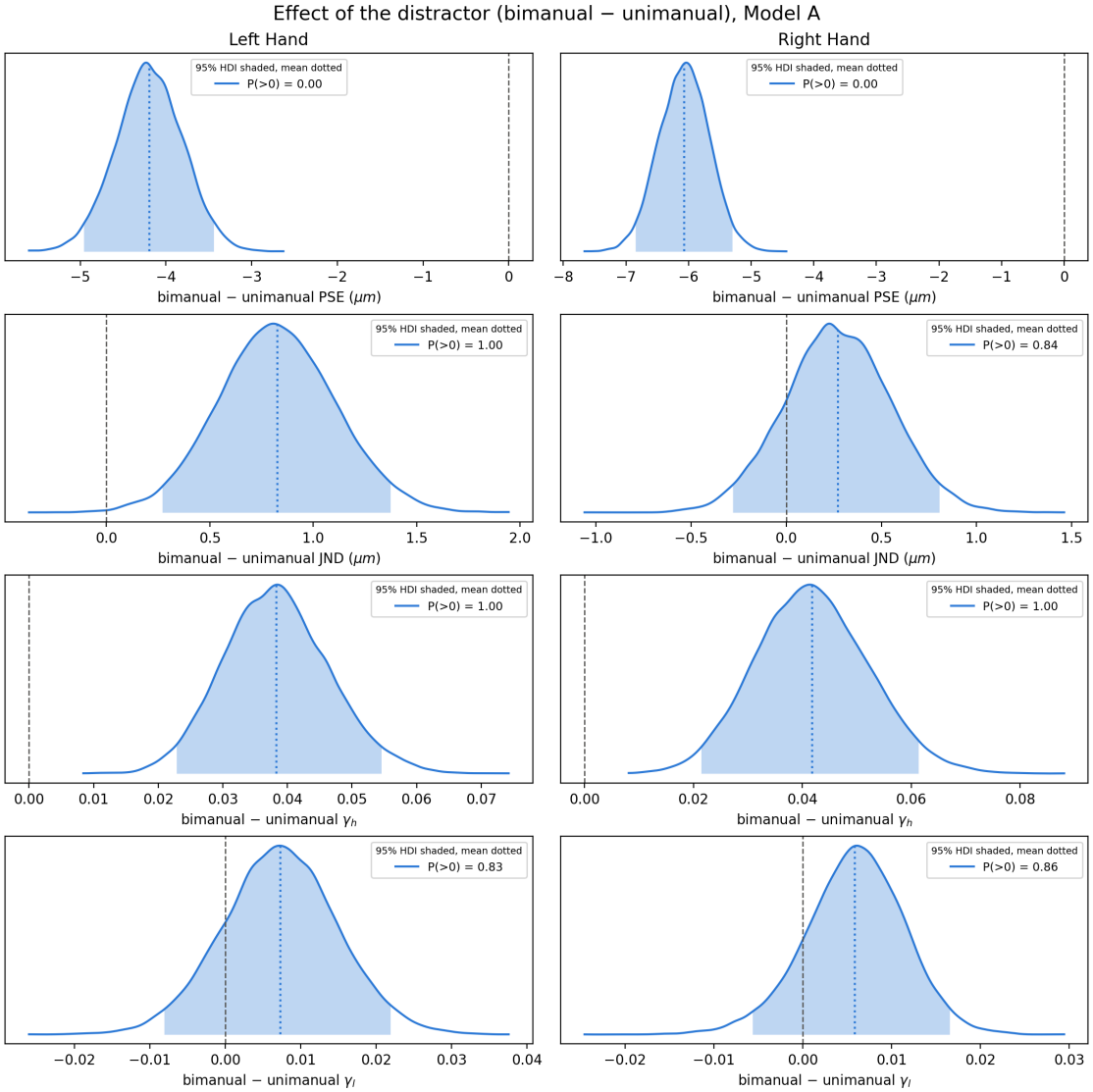


Figure 5: Posterior of the distractor effect (bimanual − unimanual), with the 95% HDI shaded and the posterior probability that the effect is positive.

The distractor shifts the PSE down by 4–6 μm and raises γ_h by about 0.04, both with probability essentially 1 (Table 2, Figures 4 and 5): with a distractor on the other hand the monkey reports "high" at lower amplitudes, and occasionally even for the weakest stimuli. The JND rises by 0.83 μm for the left hand ($P > 0$: 0.998); for the right hand the rise of 0.27 μm is uncertain ($P > 0$: 0.84). Any effect on γ_l is small and uncertain.

5 Model Checks and Comparison with Model B

5.1 Posterior predictive checks

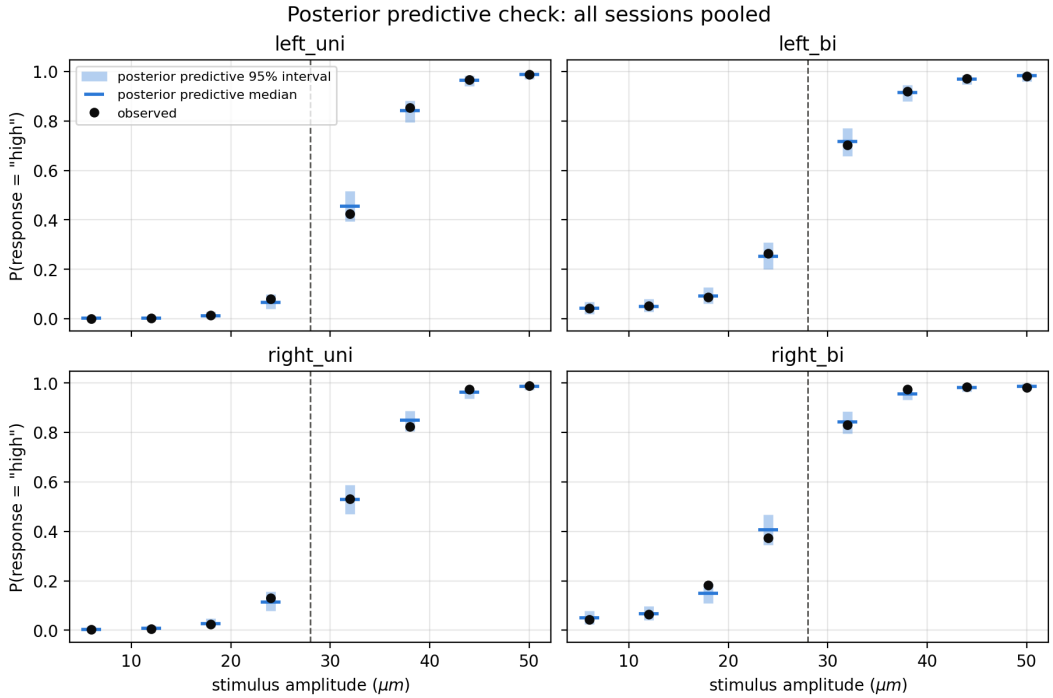


Figure 6: Observed proportion of "high" responses at each amplitude, pooled over sessions (black), against the posterior predictive median and 95% interval (blue). The dashed line is the trained threshold.

Pooled over sessions, all 32 group-by-amplitude proportions lie inside their 95% posterior predictive intervals (Figure 6): one curve per group describes the pooled data well.

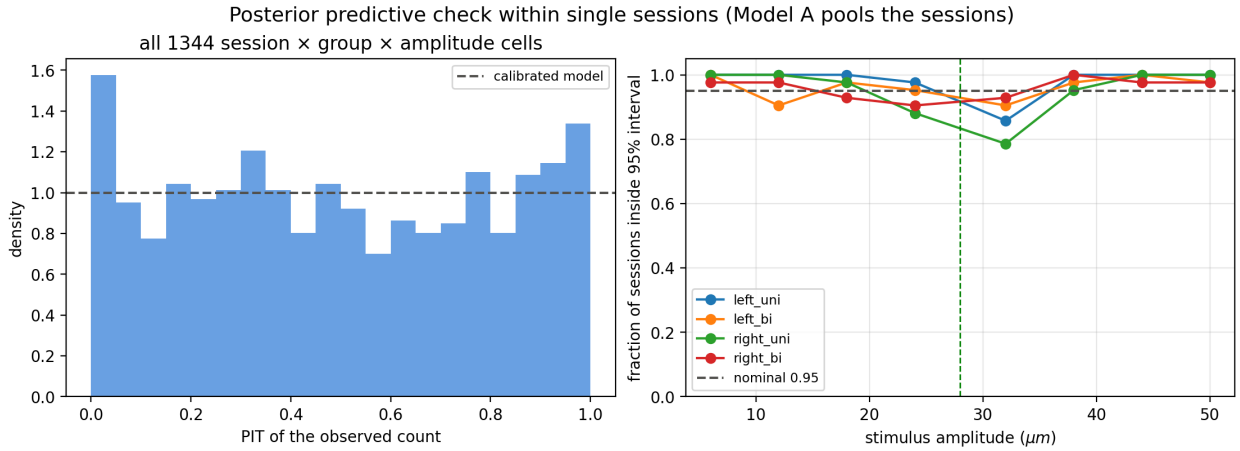


Figure 7: Posterior predictive check within single sessions. Left: PIT values of the observed number of "high" responses in each session × group × amplitude cell (flat at 1 for a calibrated model). Right: fraction of the 42 sessions whose observed count lies inside its 95% predictive interval, by group and amplitude.

Within single sessions the picture is different (Figure 7). Model A predicts the same curve in every session, so it treats the variation between sessions as if it were trial noise. Overall, 1,293 of the 1,344 session × group × amplitude cells (96.2%) lie inside their 95% intervals, close to nominal, but 4.5% of the PIT values are below 0.025 and 4.0% above 0.975, against 2.5% each for a calibrated model: the session counts vary more than Model A predicts. The excess is concentrated at 32 μm, the level just above the threshold, where the curve is steepest and a small shift of a session's PSE changes the response rate most: there only 79% (right_uni) and 86% (left_uni) of sessions lie inside their intervals. This matches the Model B fit, which found the right-hand unimanual PSE drifting upward over the study. For comparison, Model B, which gives each session its own curve, has 99.8% of these cells inside their intervals.

5.2 Leave-one-out cross-validation

Model A has $\mathrm{elpd}_{\mathrm{loo}} = -3344.0$ (SE 64.5) with $p_{\mathrm{loo}} = 13.4$ and every Pareto k at most 0.07. Model B has -3180.0 (SE 64.3) with $p_{\mathrm{loo}} = 179$. Both were fit to the same trials in the same order, so the difference can be computed trial by trial: Model B is better by 164.1 (SE 19.2), more than eight standard errors. Model B's 179 effective parameters against Model A's 13 are more than paid for by its better predictions: the sessions really differ.

5.3 Comparison of the estimates

Table 3: Posterior means of the group quantities: Model A (sessions pooled) against the average of Model B's 42 session estimates.

	PSE (μm)		JND (μm)		γ_h		γ_l	
	A	B	A	B	A	B	A	B
left_uni	32.58	32.45	3.56	3.31	0.002	0.002	0.009	0.010
left_bi	28.38	28.16	4.39	4.34	0.041	0.027	0.016	0.010
right_uni	31.57	31.44	4.09	3.57	0.003	0.003	0.006	0.008
right_bi	25.50	25.38	4.36	4.06	0.045	0.038	0.012	0.011

The PSEs agree to within 0.22 μm (Table 3). The JNDs of Model A are larger, by 0.05–0.52 μm, most for right_uni. This is what pooling does: averaging curves whose PSEs differ from session to session gives a shallower curve than any single session has, so the pooled JND mixes the monkey's precision within a session with the drift of its PSE between sessions. Model B separates the two. Model A's bimanual γ_h is also higher (0.041 against 0.027 for left_bi), probably for a related reason: in sessions whose PSE is unusually low, "high" responses at the weakest stimuli are more common, and the pooled curve can only absorb them as lapses.

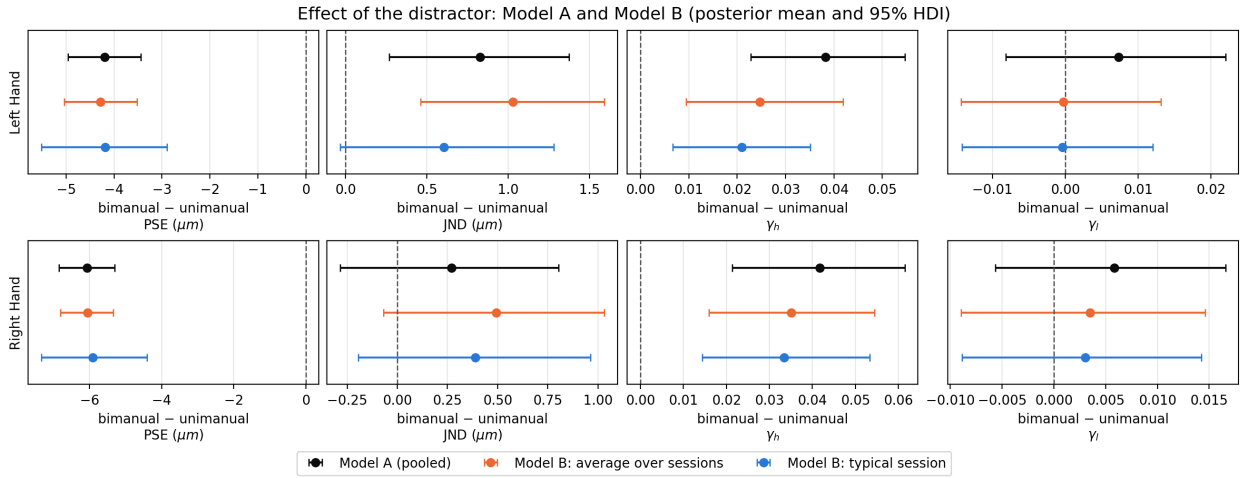


Figure 8: Distractor effect under Model A (black), and under Model B averaged over the 42 sessions (orange) and for a typical session (blue): posterior mean and 95% HDI.

The distractor effects agree well (Figure 8). The PSE shift is −4.19 (A) against −4.29 (B, average over sessions) for the left hand and −6.07 against −6.06 for the right hand, with HDIs of nearly the same width (about 1.5 μm). The γ_h effect is somewhat larger under Model A (0.038 against 0.025 for the left hand), and the JND effect somewhat smaller (0.83 against 1.03 for the left hand, 0.27 against 0.49 for the right), but the HDIs overlap widely and the conclusions are the same.

The difference between the models lies in what the estimates describe. Model A's effects, like Model B's average over sessions, describe these 42 sessions. For a new session from the same population, Model B's typical-session effect is the relevant one, and its HDI is about twice as wide

for the PSE (2.6 and 2.9 μm against 1.5): the uncertainty about how sessions vary, which Model A cannot express.

6 Conclusions

- The fit is reliable: no divergences, $\hat{R} = 1.000$, bulk ESS above 15,000, and the pooled data are reproduced at every amplitude.
- The distractor lowers the PSE by 4.2 (left) and 6.1 (right) μm and raises γ_h by about 0.04; it raises the left-hand JND by about 0.8 μm. These conclusions agree with Model B.
- Model A ignores the variation between sessions, which is real: Model B predicts the data much better (elpd difference 164, SE 19), and within single sessions the responses near the threshold vary more than Model A allows. As a result Model A's JNDs and bimanual γ_h are somewhat inflated, and its HDIs describe only the sessions observed.
- Model A is a useful, fast summary of the pooled data and a check on Model B, but Model B should be used for inference.

A Further Figures

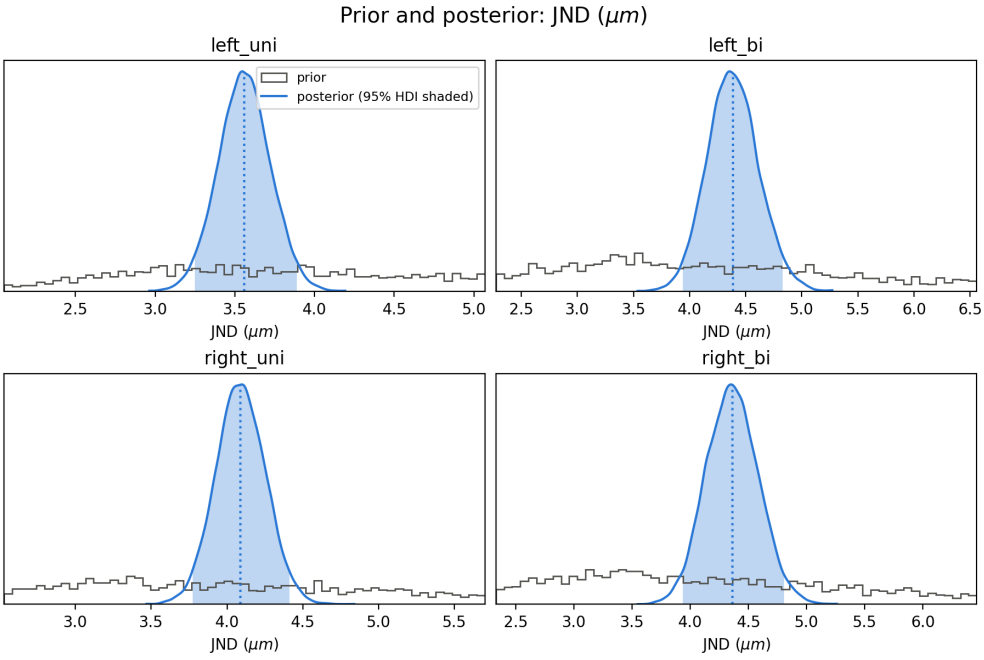


Figure 9: Prior (gray) and posterior (blue) of the JND.

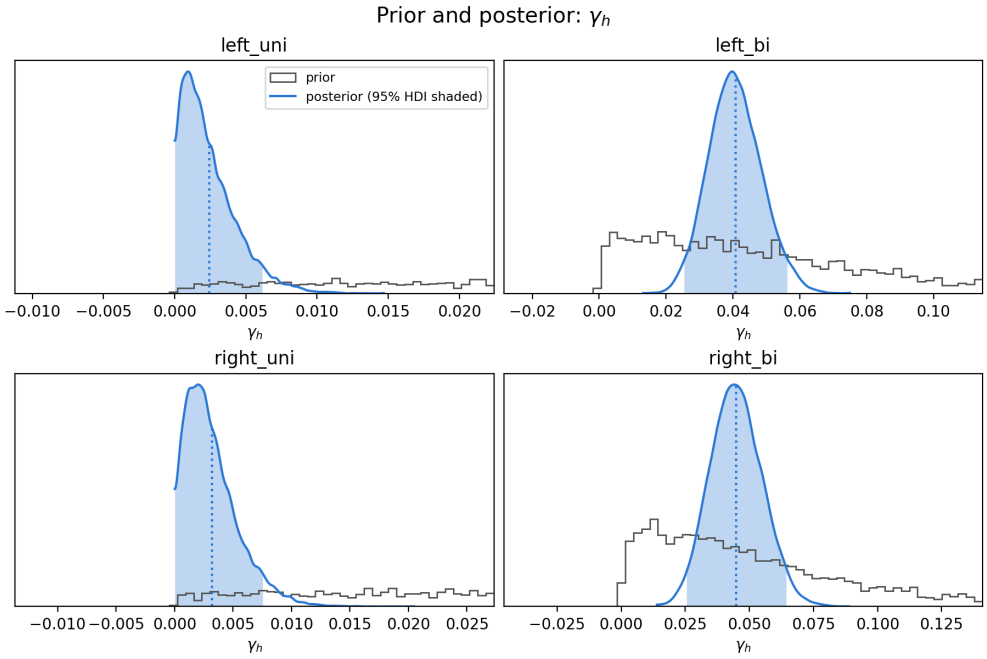


Figure 10: Prior (gray) and posterior (blue) of γ_h.

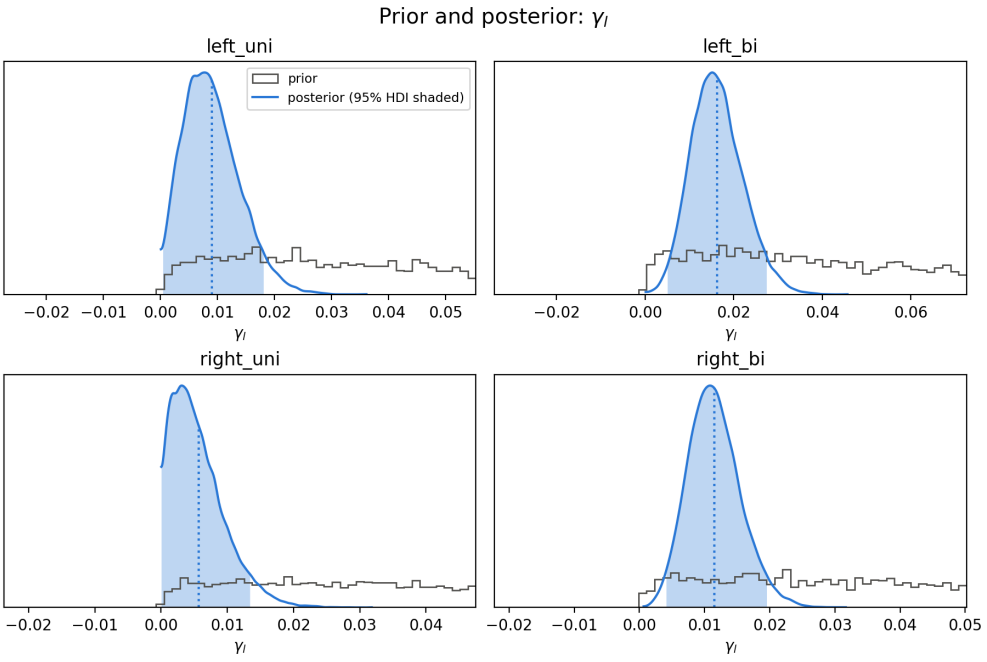


Figure 11: Prior (gray) and posterior (blue) of γ_l.

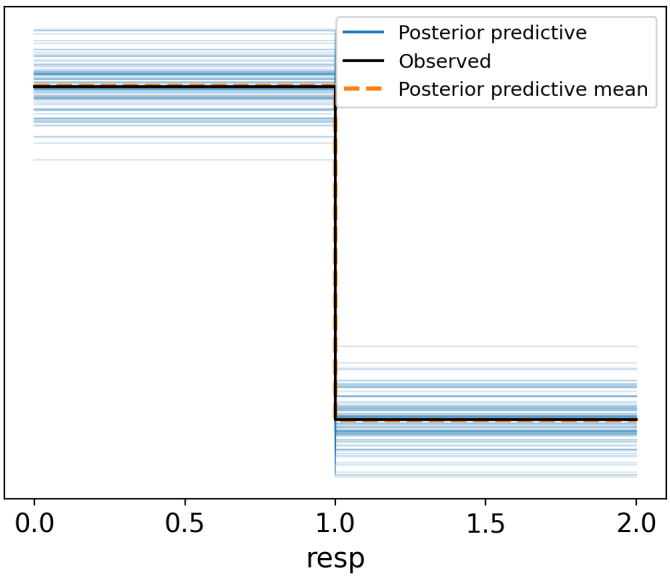


Figure 12: ArviZ posterior predictive check of the responses (0 = “low”, 1 = “high”), pooled over all trials: observed (black), 100 posterior predictive draws (blue) and their mean (orange dashed).

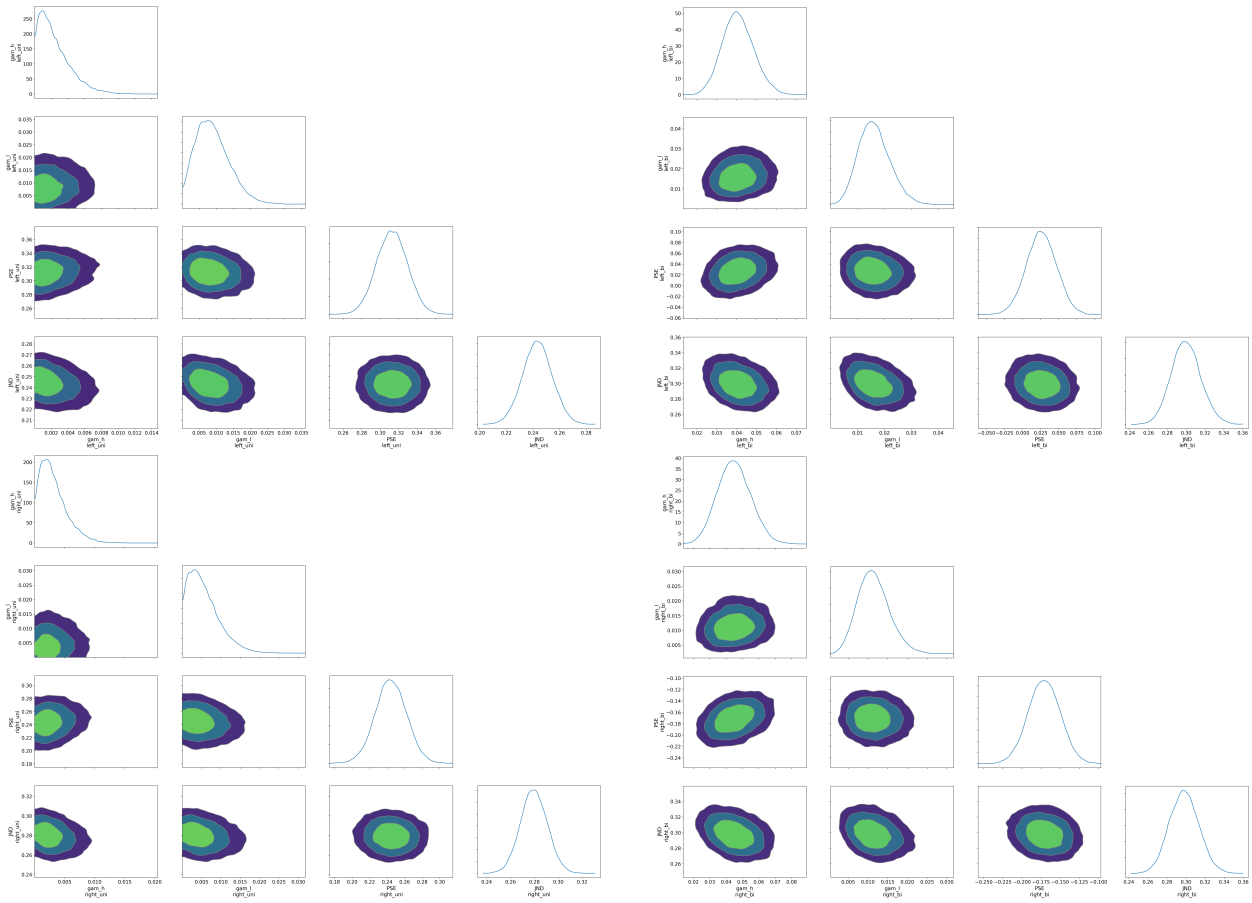


Figure 13: Joint posteriors of γ_h, γ_l, PSE and JND (standardized units) for left_uni, left_bi (top) and right_uni, right_bi (bottom).